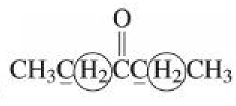
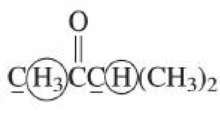
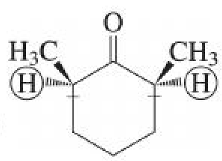
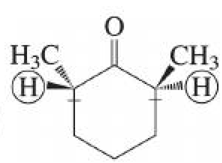
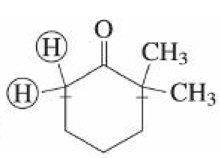
Solutions to Problems

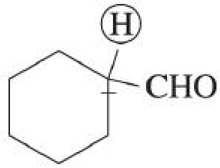
32. (a) 

(b) 

(c) 

(d) 

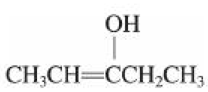
(e) 

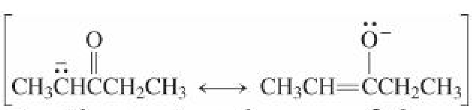
(f) 

(g) 

(h) 

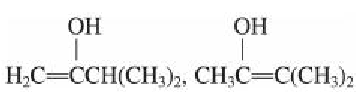
33. (a)

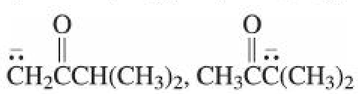
(i) 

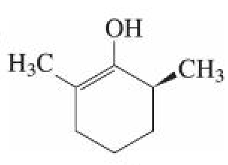
(ii) 

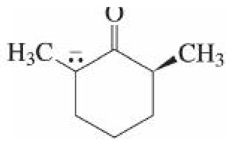
For the rest, only one of the enolate resonance forms will be written.

(b)

(i) 

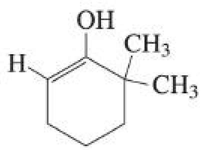
(ii) 

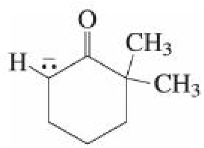
(c) (i) 

(ii) 

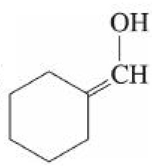
(d) Same as (c). Stereochemistry will be lost as α-carbonα-carbon becomes sp2.sp^2 .

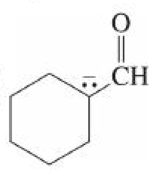
(e)

(i) 

(ii) 

(f)

(i) 

(ii) 

(g) None possible (no α-hydrogens!)α-hydrogens!)

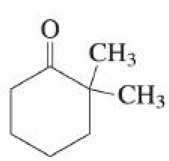
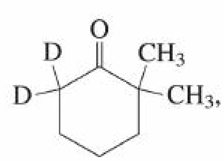
(h)

(i) $(CH_3)_3CCH{=}\overset{OH}{\overset{|}{C}}H$

(ii) $(CH_3)_3C\overset{-}{\ddot{C}}H\overset{O}{\overset{\|}{C}}H$

34. **(a)** Replace **all** α-hydrogensα-hydrogens with D;D; for instance,

$CH_3CH_2\overset{O}{\overset{\|}{C}}CH_2CH_3$ gives $CH_3CD_2\overset{O}{\overset{\|}{C}}CD_2CH_3$,

 gives  and

$(CH_3)_3CCH_2\overset{\overset{O}{\|}}{C}H$ gives $(CH_3)CCD_2\overset{\overset{O}{\|}}{C}H$.

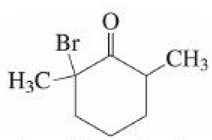
(Note how the aldehyde hydrogen is **not** replaced—it is **not** acidic.)

(b)

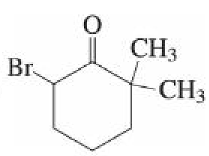
Conditions for introduction of a **single** α-halogen.α-halogen. In order, the products are

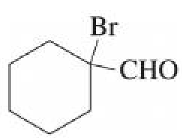
1(a) $CH_3CHBr\overset{\overset{O}{\|}}{C}CH_2CH_3$

1(b) A mixture of $CH_3\overset{\overset{O}{\|}}{C}CBr(CH_3)_2$ and $BrCH_2\overset{\overset{O}{\|}}{C}CH(CH_3)_2$

1(c), (d) 

(Mixture of stereoisomers from either cis or trans starting ketone)

1(e) 

1(f) 

1(g) No reaction

1(h) $(CH_3)_3CCHBr\overset{\overset{O}{\|}}{C}H$

(c) All α-hydrogensα-hydrogens are replaced by ClCl under these conditions.

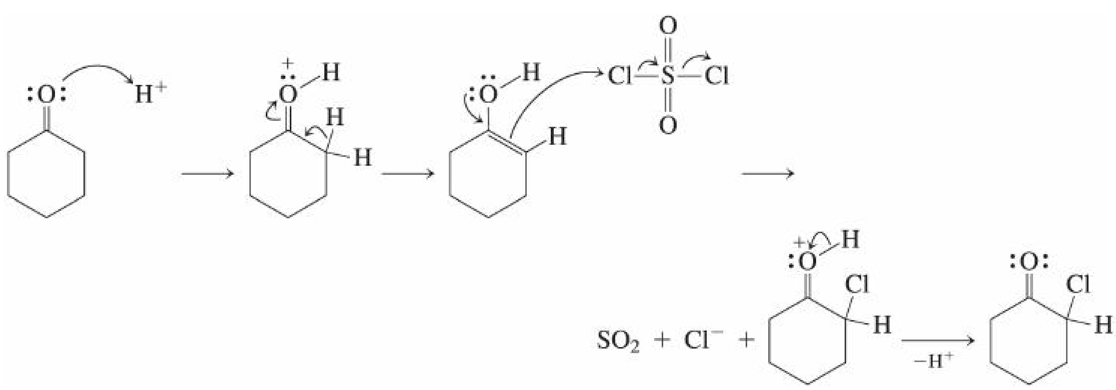
35. **(a)** An equivalent of Br2Br_2 in acetic acid (CH3CO2H)(CH_3CO_2H) solvent

(b) Excess Cl2Cl_2 in aqueous base

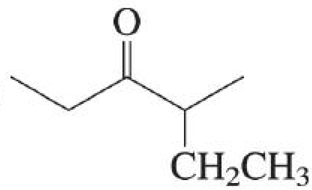
(c) One equivalent of Cl2Cl_2 in acetic acid

(d) Excess Br2Br_2 in aqueous base

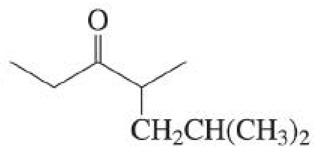
36. Conditions are acidic; use only neutral or positively charged organic structures. (The negatively charged chloride ion leaving group gets a pass: It is too weak a base to be protonated under conditions of catalytic acid.)



37.

(a) 

(b) 3-Pentanone+H2C=CHCH3 3-Pentanone + $H_2C{=}CHCH_3$

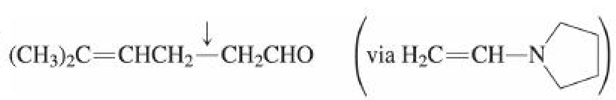
(c) 

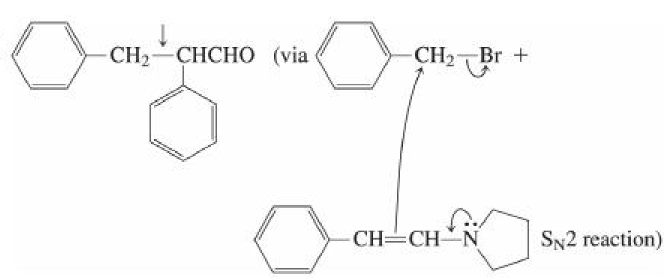
(d) 3-Pentanone+H2C═C(CH3)2
3-Pentanone + H_2CâC$(CH_3)_2$

(b) and (d) show the results of E2 elimination: Alkylation requires an SN2S_N2 process and you already know that these are poor with secondary and impossible with tertiary haloalkanes.

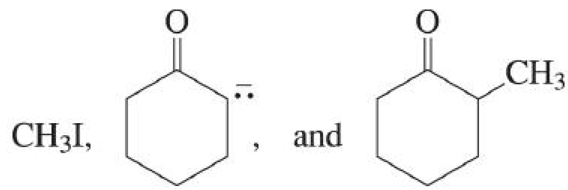
38. Both are aldehyde→enamine→alkylation sequences.

The new carbon–carbon bond is marked with an arrow.

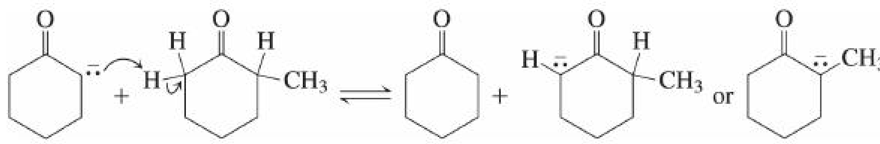
(a) 

(b) 

39. Illustrated with cyclohexanone: Before the reaction between cyclohexanone enolate and iodomethane has gone to completion, a mixture is present containing



An acid-base reaction between cyclohexanone enolate and 2-methylcyclohexanone can take place under these conditions, leading to either possible enolate of the latter.

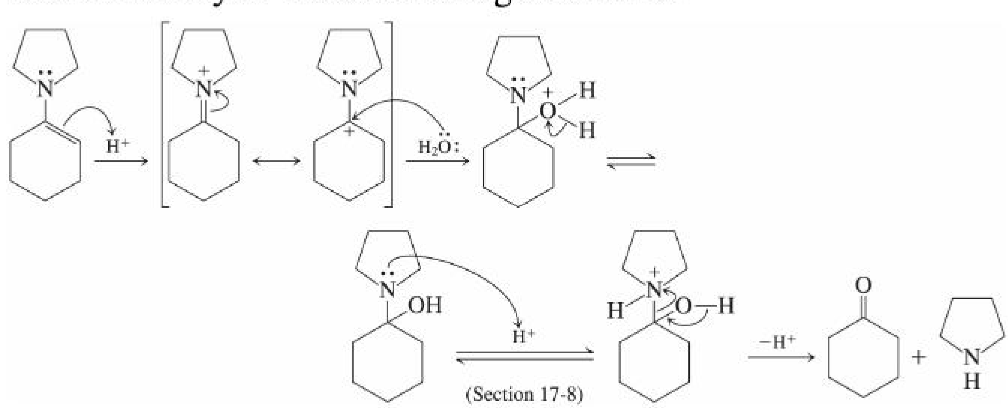


Reaction of the new enolates with CH3ICH_3I leads to double alkylation products.

Enamines reduce this problem. They are less reactive and more selective than enolate ions.

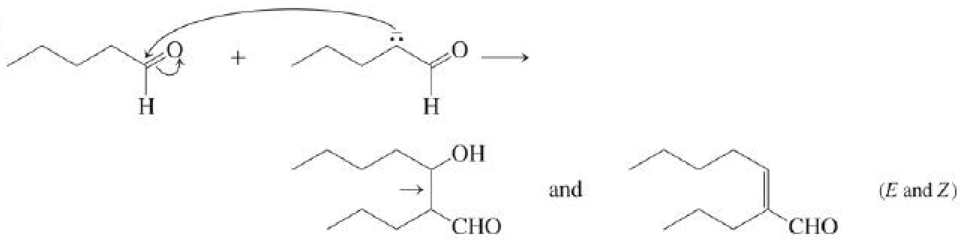
40. Yes. Enamines (neutral) are much less basic than enolates (anionic) and show much less tendency to cause E2 elimination reactions.

41. Use the catalyst! Then it's straightforward:

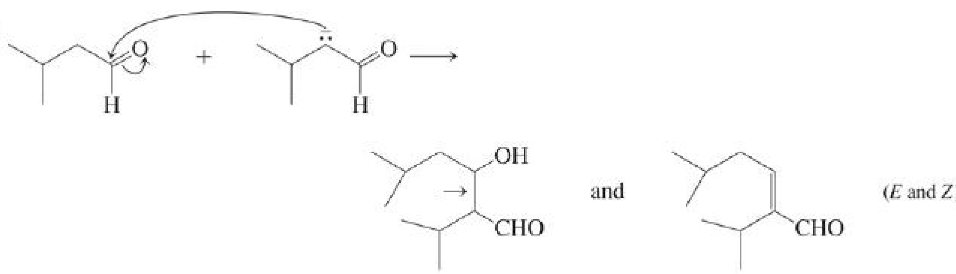


42. The direction of nucleophilic attack is shown, and the new bond is marked in the product with an arrow. Both the initial hydroxycarbonyl compound and the enone that forms upon dehydration are shown.

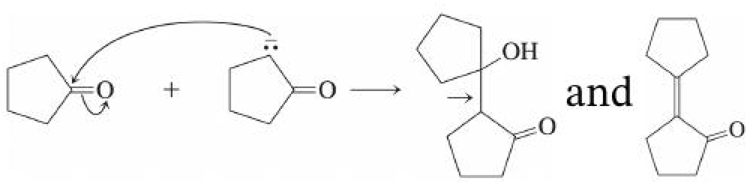
(a)



(b)

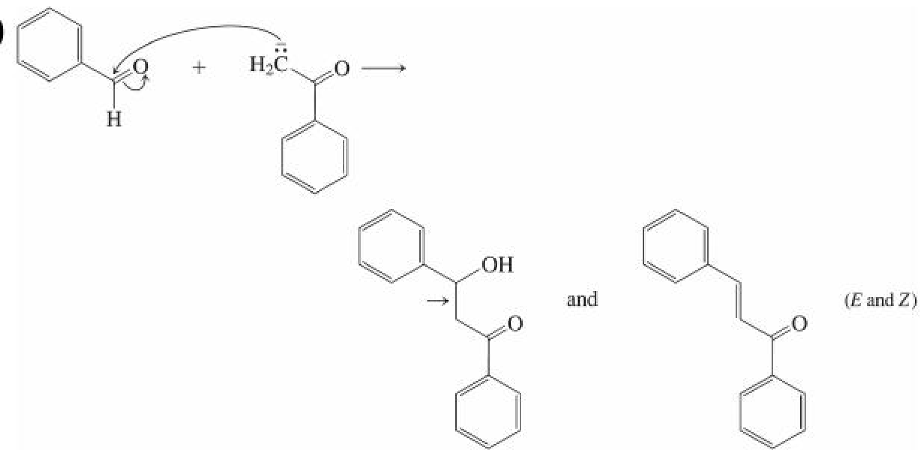


(c)



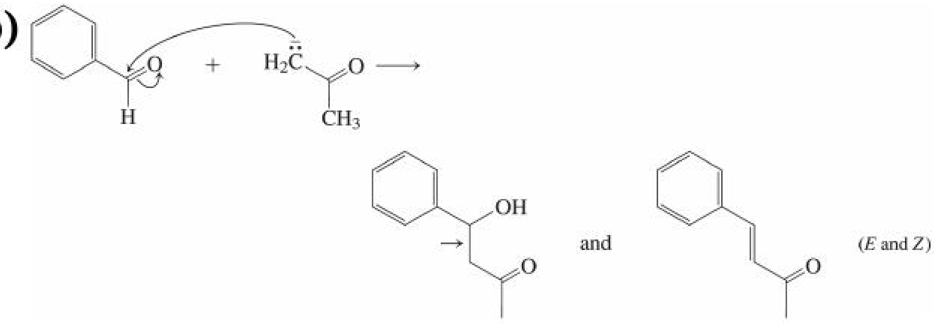
13. The direction of nucleophilic attack is shown, and the new bond is marked in the product with an arrow. Both the initial hydroxycarbonyl compound and the enone that forms upon dehydration are shown.

(a)



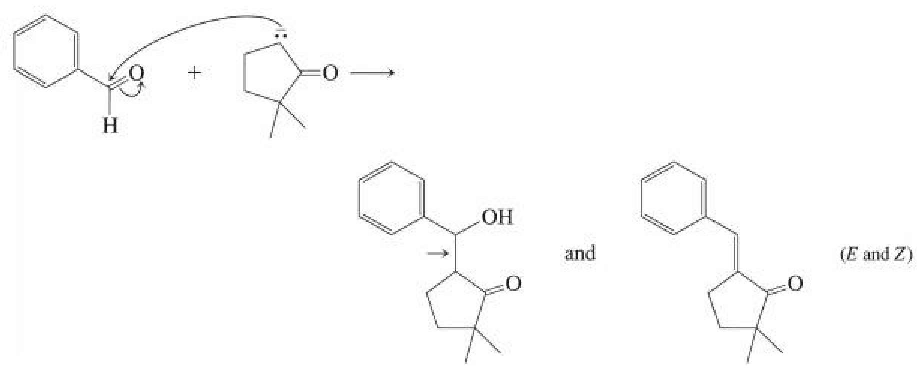
H
$H_2\ddot{C}$
O
OH
O
and
O
(*E* and *Z*)

(b)

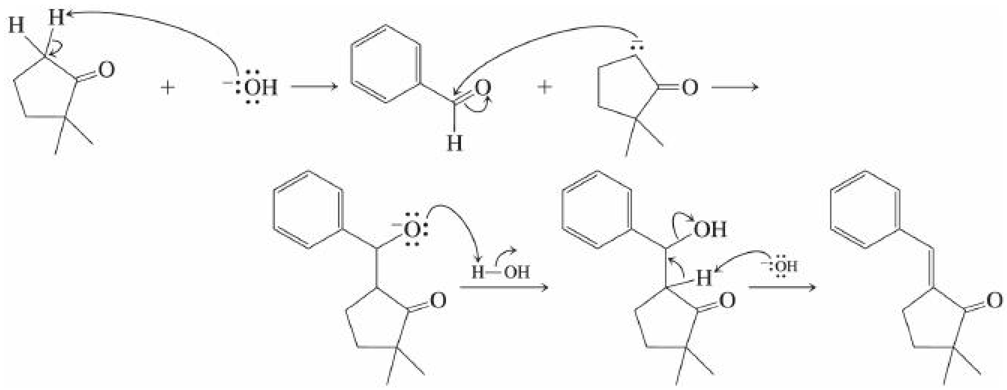


H
$H_2\ddot{C}$
O
CH_3
OH
O
and
O
(*E* and *Z*)

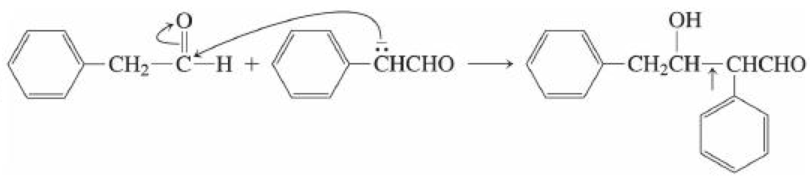
(c)

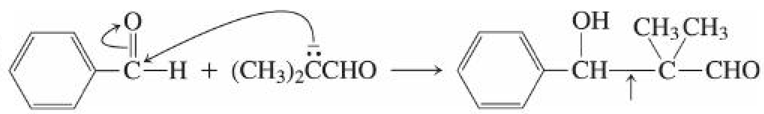


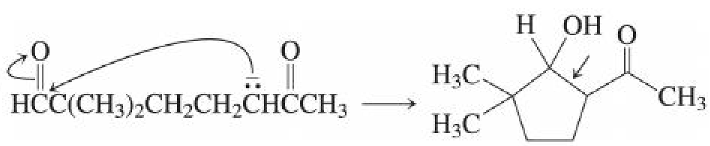
H
O
OH
O
and
O
(*E* and *Z*)

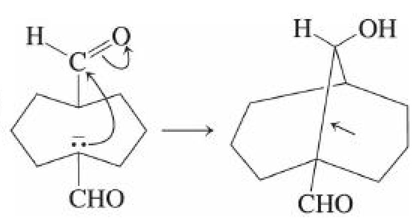
14. 

15. Aldol additions. The direction of nucleophilic attack is shown, and the new bond is marked in the product with an arrow.

(a) 

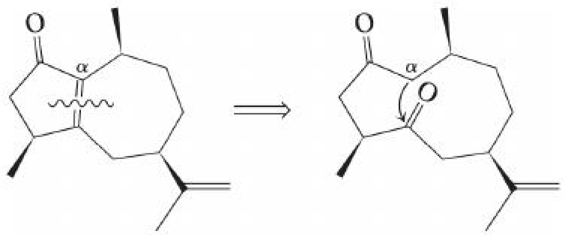
(b) 

(c) 

(d) 

16. This problem is essentially an exercise in retrosynthetic analysis. The intramolecular aldol condensation forms the C=C Câ C double bond of an α,β-α,β-unsaturated carbonyl compound by linking the α carbon α carbon of one carbonyl function to a second carbonyl carbon five or

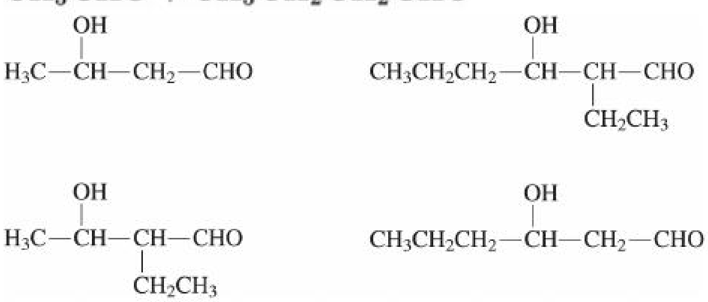
six carbon atoms away in the same molecule. Following that recipe, we solve the problem as shown, with the curved arrow showing the pathway for formation of the new carbon-carbon bond:



Rotundone

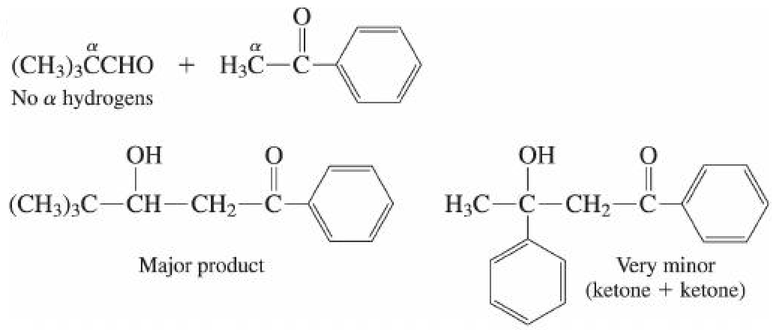
47 and 48. (a) CαH3CHO+CH3CH2CαH2CHO

$\overset{\alpha}{C}H_3CHO + CH_3CH_2\overset{\alpha}{C}H_2CHO$

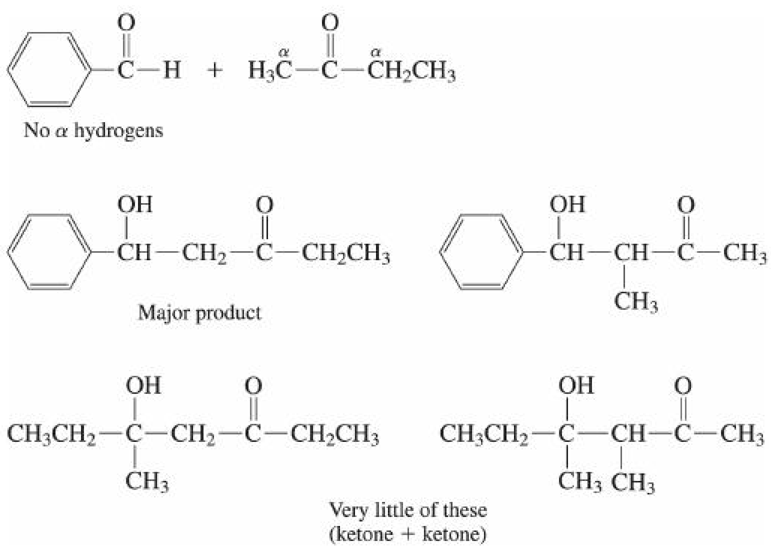


None of these compounds will be a "major" product, but the first one may form in somewhat higher yield than the others because it is the least sterically crowded.

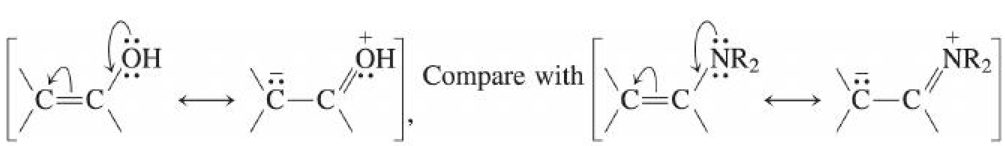
(b)



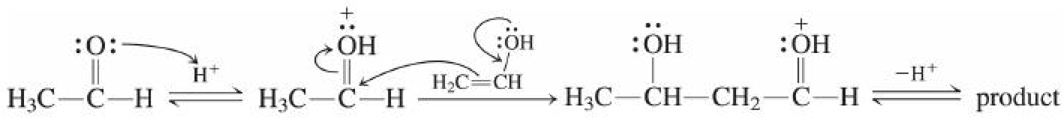
(c)



49. Because enolates are not feasible intermediates, consider an alternative—a neutral enol, which should be a nucleophile somewhat similar to an enamine.



How can acid play a role? You already know (Section 17-5) that acid can catalyze carbonyl addition reactions by attachment to the carbonyl oxygen, thereby generating a better electrophile. You've also seen that acid catalyzes conversion of a carbonyl compound into the corresponding enol (Section 18-2). So (using acetaldehyde for illustration),



50.

(a) Cl2,CH3CO2H,H2O;Cl_2, CH_3CO_2H, H_2O;

(b) Br2,NaOH,H2O;$Br_2, NaOH, H_2O$;

(c) 1. R2NH,H+,R_2NH, H^+, 2. CH3I;CH_3I;

(d) 1. R2NH,H+,R_2NH, H^+, 2. C6H5CH2Br;$C_6H_5CH_2Br$;

(e) 1. R2NH,H+,R_2NH, H^+, 2. H2C=CHCH2Br;
H_2Câ $CHCH_2Br$;

(f) NaOH,H2O,$NaOH, H_2O$, 5°C$5^\circ C$ (aldol reaction);

(g) NaOH,H2O,$NaOH, H_2O$, heat (aldol condensation);

(h) benzophenone (excess), NaOH,H2O,$NaOH, H_2O$, heat (crossed aldol condensation).

51.

(a) 1. R2NH,H+,R_2NH, H^+, 2. CH3CH2Br;CH_3CH_2Br;

(b) Cl2,NaOH,H2O;$Cl_2, NaOH, H_2O$;

(c) CH3CH2CHOCH_3CH_2CHO (excess), NaOH,H2O, $NaOH, H_2O$, 5°5° (crossed aldol condensation);

(d) 1. R2NH,H+,R_2NH, H^+, 2. H2C═CHCH2Br; H_2Câ $CHCH_2Br$;

(e) CH3CH2CHOCH_3CH_2CHO (excess), NaOH,H2O, $NaOH, H_2O$, heat (crossed aldol condensation);

(f) Br2,CH3CO2H,H2O.Br_2, CH_3CO_2H, H_2O.

52.

(a) 1. (C6H5)2CuLi,$(C_6H_5)_2CuLi$, 2. H+;H^+ ;

(b) 1. (CH3)2CuLi,$(CH_3)_2CuLi$, 2. H+;H^+ ;

(c) H2,Pd;H_2, Pd;

(d) 1. (CH3)2CuLi,$(CH_3)_2CuLi$, 2. CH3I;CH_3I;

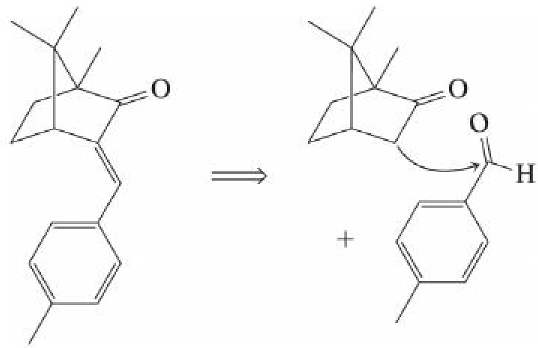
(e) LiAlH4;$LiAlH_4$;

(f) H2O,−OH;$H_2O, {}^-OH$;

(g) CH3MgI;CH_3MgI;

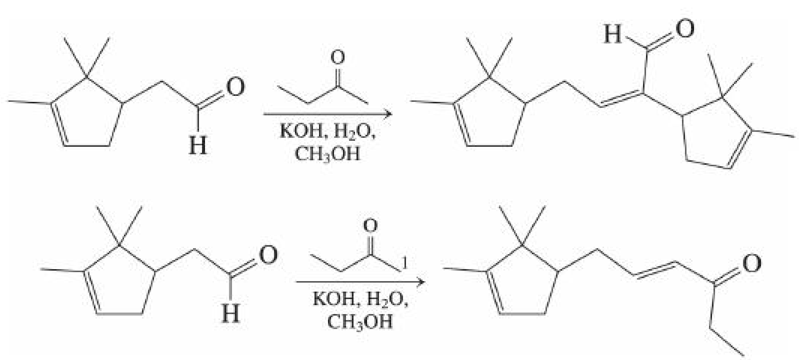
(h) cyclohexanone (excess), NaOCH2CH3,CH3CH2OH $NaOCH_2CH_3, CH_3CH_2OH$ (conjugate addition [Michael addition] of the enolate of cyclohexanone).

53. **(a)** Retrosynthetically,



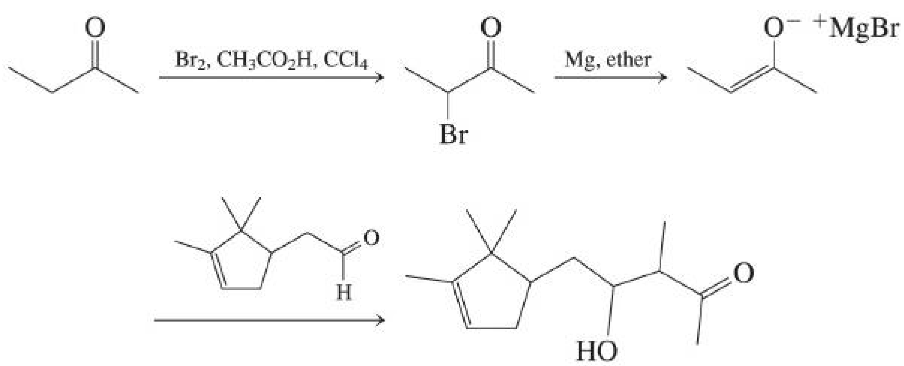
(b) Both compounds constitute extended conjugated systems that may absorb UV light. 4-MBC's ketone function is conjugated with a double bond and its attached benzene ring. Gadusol's carbonyl is also conjugated with an alkene double bond. In addition, the lone pairs of the hydroxy oxygen at the ββ position may delocalize into the enone system.

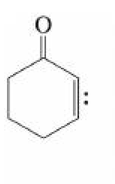
54. **(a)** The desired reaction is a crossed (or mixed) aldol condensation, but one that suffers from competition with at least two other aldol processes: condensation between two molecules of aldehyde, and condensation of the aldehyde with the methyl group (C1)(C1) of the ketone:



In addition, both of these processes, as well as the condensation that affords the desired compound, give a mixture of *E* and *Z* stereoisomers.

(b) This solution forces the formation of just a single nucleophilic enolate carbon, C3C3 of the ketone. The side reactions described above derive from reaction of nucleophilic enolate ions at C2C2 of the aldehyde and C1C1 of the ketone, respectively. By restricting nucleophilic behavior to just the single carbon atom required to allow access to the desired product structure, the two side reactions above are eliminated. In addition, dehydration of the product of this sequence can be controlled to give the necessary stereoisomer, because that isomer places the two larger substituents on the side-chain double bond trans to each other, making it the more stable and the likely product of dehydration under equilibrium (thermodynamic) conditions.

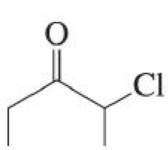


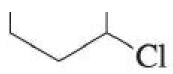
55. 

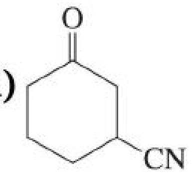
:

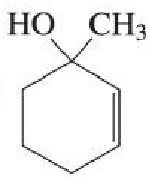
(a) Cyclohexanone

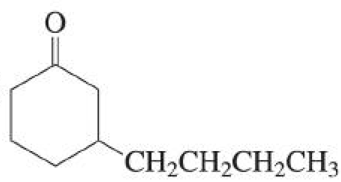
(b) 2-Cyclohexenol

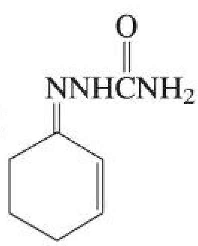
(c) 

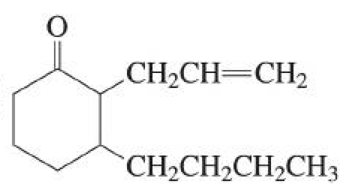


(d) 

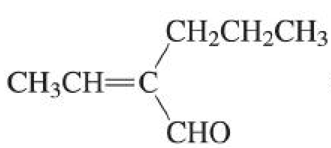
(e) 

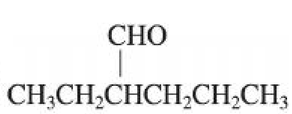
(f) 

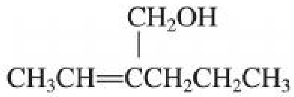
(g) 

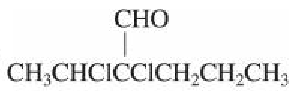
(h) 

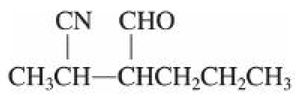
(first step forms an enolate, which is alkylated)

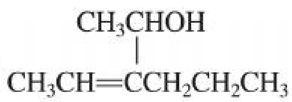
 :

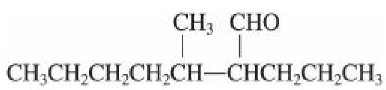
(a) 

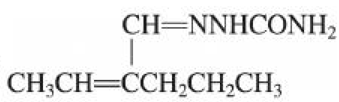
(b) 

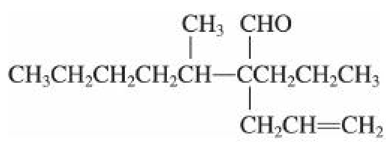
(c) 

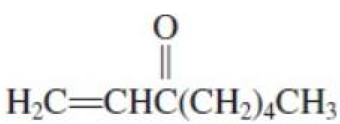
(d) 

(e) 

(f) 

(g) 

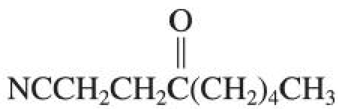
(h) 

 :

(a) 3-Octanone

(b) 1-Octen-3-ol

(c) 

(d) 

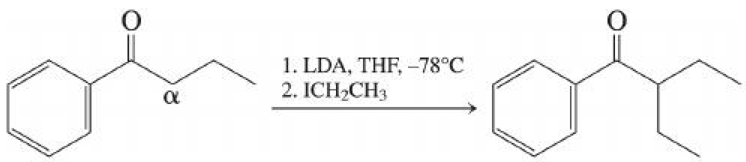
(e) $H_2C{=}CH\overset{\displaystyle OH}{\underset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}(CH_2)_4CH_3$

(f) $CH_3(CH_2)_5\overset{\displaystyle O}{\overset{\|}{C}}(CH_2)_4CH_3$

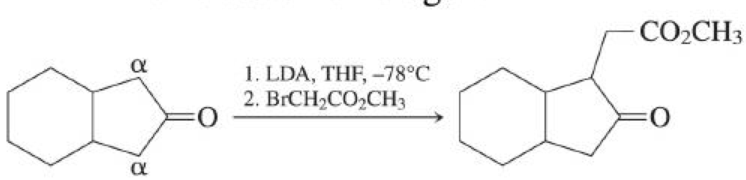
(g) $H_2C{=}CH\overset{\displaystyle NNHCONH_2}{\overset{\|}{C}}(CH_2)_4CH_3$

(h) $CH_3(CH_2)_4\underset{\displaystyle CH_2CH{=}CH_2}{\underset{|}{C}}H\overset{\displaystyle O}{\overset{\|}{C}}(CH_2)_4CH_3$

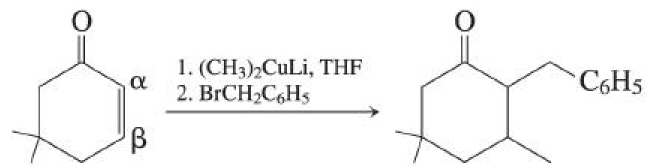
56. **(a)** Making a bond at an α-carbonα-carbon implies alkylation: Deprotonation by a strong base to form the enolate, followed by reaction with a haloalkane. The substrate has only one α-carbon, α-carbon, so the choice of base is not particularly critical. Most chemists would use LDA (see Exercise 18-2).



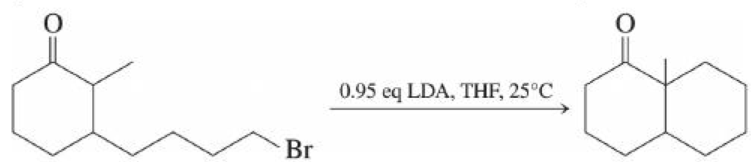
(b) A similar situation. Here, the two α-carbonsα-carbons are equivalent, so where deprotonation occurs is irrelevant. Do you see how to reconstruct the necessary alkylating agent from the structure of the desired product? Disconnect it from the α-carbon α-carbon and attach a halogen.



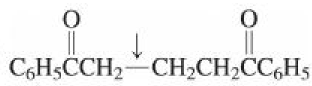
(c) Now bond formation is required at both the α- and β-carbons. α– and β-carbons. We have a model for this process in Section 18-10: conjugate addition of a cuprate reagent to an α,β-α,β-unsaturated ketone, followed by alkylation of the resulting enolate.

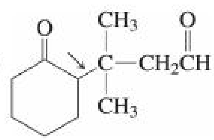


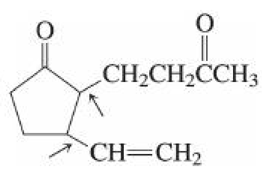
(d) It is now an intramolecular bond construction at an α-carbon α-carbon that we wish to effect. In order to give the enolate at the required (more-substituted) position, we may use LDA at room temperature in the presence of a slight excess of ketone as a proton source to facilitate enolate equilibration.

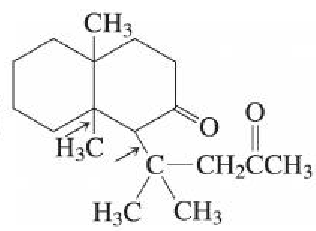


57. Michael additions: 1,4-additions of enolate ions to α,β-α,β-unsaturated aldehydes or ketones. New bond(s) indicated by arrow(s).

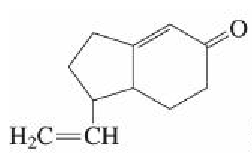
(a) 

(b) 

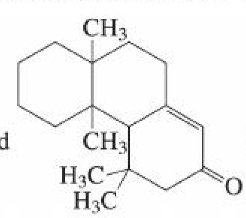
(c) 

(d) 

(e) Intramolecular aldol condensations, leading to



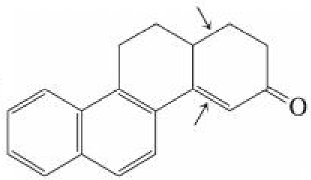
from (c) and



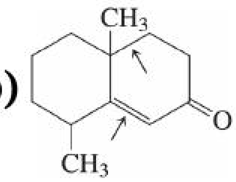
from (d)

58. Robinson annulation sequences (Michael addition followed by aldol).

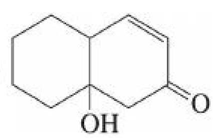
(a)



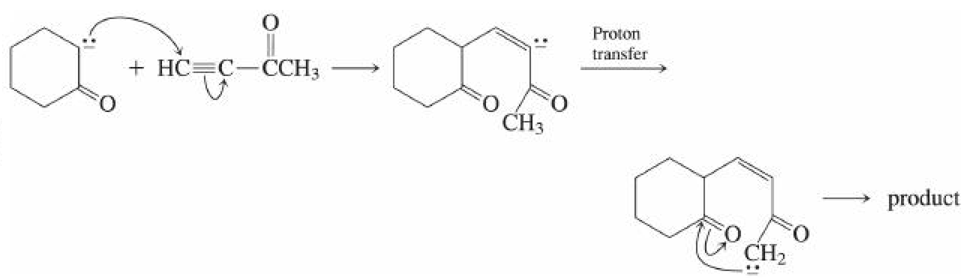
(b)



(c)

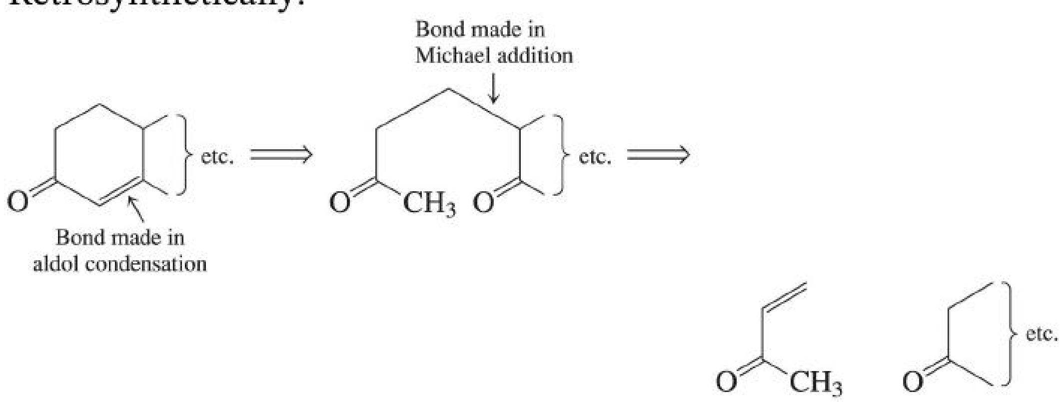


(d)

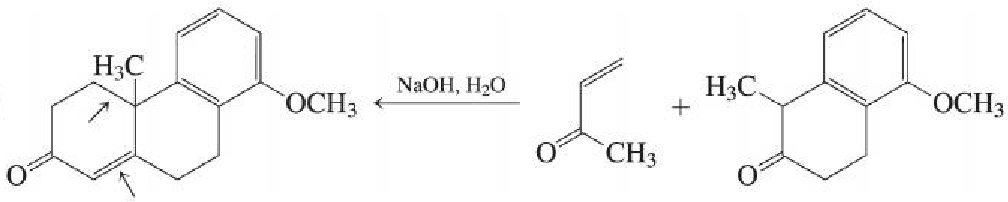


Heating in (a) and (b) tends to drive dehydration to give the α,β-α,β- unsaturated ketone product.

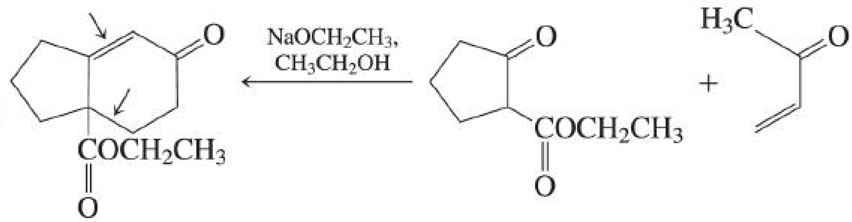
59. Retrosynthetically:



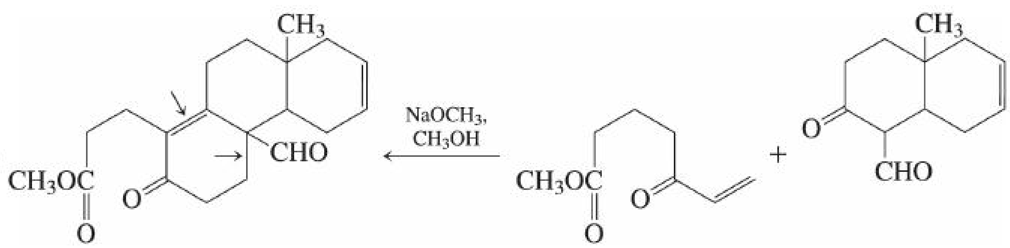
(a)



(b)

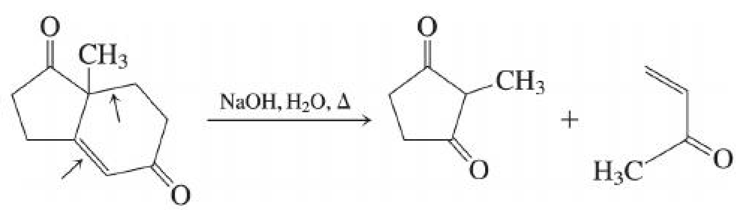


(c)

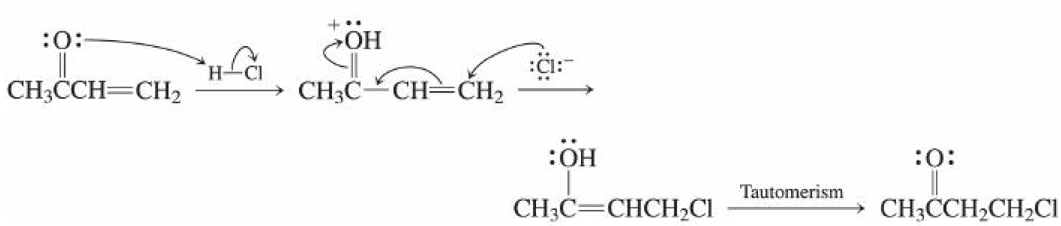


It’s not a big deal, but use of NaOH,H2O$\mathrm{NaOH, H_2O}$ in (b) and (c) would hydrolyze the ester groups to carboxylic acids (—CO2H).$(\mathrm{—CO_2H})$. Using NaOR,ROH$\mathrm{NaOR, ROH}$ where RR corresponds to the alkyl group of the ester, prevents this (Chapter 20).

(d)



50. No! The carbonyl group changes the mechanism.



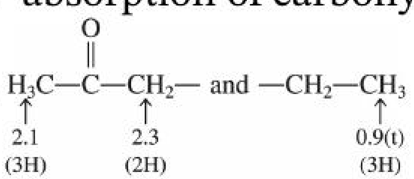
Protonation occurs on oxygen, not carbon, and the result **looks** like it's anti-Markovnikov. It is really 1,4-addition.

61. Calculate your degree of unsaturation first! The 13C^{13}C information may help get you to the answers faster, but it is not essential in these cases.

(a) Hsat=10+2=12;degree of unsaturation=12−102=1 π bond or ring.

$H_{sat} = 10 + 2 = 12$; degree of unsaturation $= \frac{12-10}{2} = 1\ \pi$ bond or ring.

UV data: n→π*$n \rightarrow \pi^*$ absorption of carbonyl.

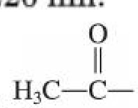
NMRNMR spectrum: 

These assignments are clear, giving as the answer 2-pentanone, CH3COCH2CH2CH3$CH_3COCH_2CH_2CH_3$ (A). The sextet at δ=1.5$\delta = 1.5$ corresponds to the C4 CH2$C4\ CH_2$ group.

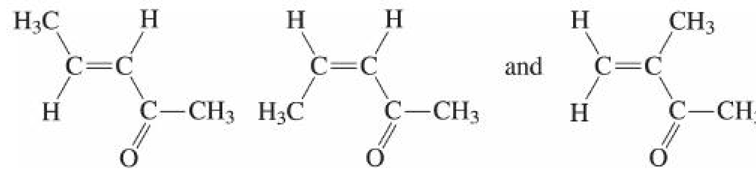
(b) Hsat=10+2=12;degree of unsaturation=12−82=2 π bonds or rings.

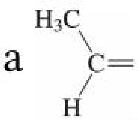
$H_{sat} = 10 + 2 = 12$; degree of unsaturation $= \frac{12-8}{2} = 2\ \pi$ bonds or rings.

UV data: π→π*$\pi \rightarrow \pi^*$ absorption of α,β-α,β- unsat'd carbonyl at 220 nm.220 nm.

NMRNMR spectrum:  is clear.

Also two alkene hydrogens (δ=5.8–6.9)($\delta = 5.8\text{–}6.9$) and another three H’sH’s at δ=1.9 (H3C—?).$\delta = 1.9\ (H_3C\text{—}?)$. Because UV indicates conjugation, there are three possibilities:

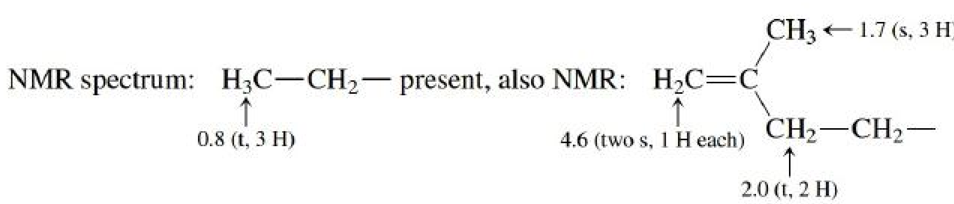


The last is ruled out because the signal at δ=1.9$\delta = 1.9$ has a large splitting, indicating that a  piece is present. In the alkene region, the large splitting of the signal at δ=6.0$\delta = 6.0$ (about 15 Hz15 Hz) suggests trans alkene H’s,H’s, so the **first** of the three possibilities above is B. Notice the doublet of quartets for the signal at δ=6.7,$\delta = 6.7$, consistent with an alkene HH split by both a second alkene HH and a methyl group.

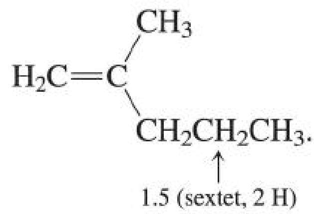
(c) Hsat=12+2=14;degrees of unsaturation=14−122=1π bond or ring.

$$H_{sat} = 12 + 2 = 14;\ \text{degrees of unsaturation} = \frac{14 - 12}{2} = 1\pi\ \text{bond or ring.}$$

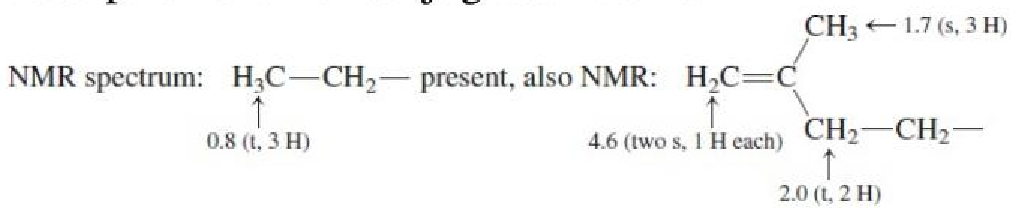
UV data: π→π*$\pi \rightarrow \pi^*$ absorption of a simple alkene.



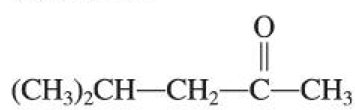
can be identified by chemical shifts and splittings. These add up to C7H14,C_7H_{14}, so one CH2CH_2 group has been duplicated in the two pieces. After correcting for that, the answer for C becomes



(d) Also 1 degree of unsaturation. UV data: n→π*n → π* absorption of a nonconjugated ketone.



All that's left to arrive at the molecular formula is CH2,CH_2, which appears as a doublet at δ=2.3;$\delta = 2.3$; insert it between the above two pieces to get the answer:



is D

The signal for the CHCH group (nine lines!) appears at the base of the CH3CH_3 singlet at δ=2.1.$\delta = 2.1$.

(e) A+H2C=P(C6H5)3→THFC $A + H_2C = P(C_6H_5)_3 \xrightarrow{\text{THF}} C$

(f) B+(CH3)2CuLi→THFD $B + (CH_3)_2CuLi \xrightarrow{\text{THF}} D$

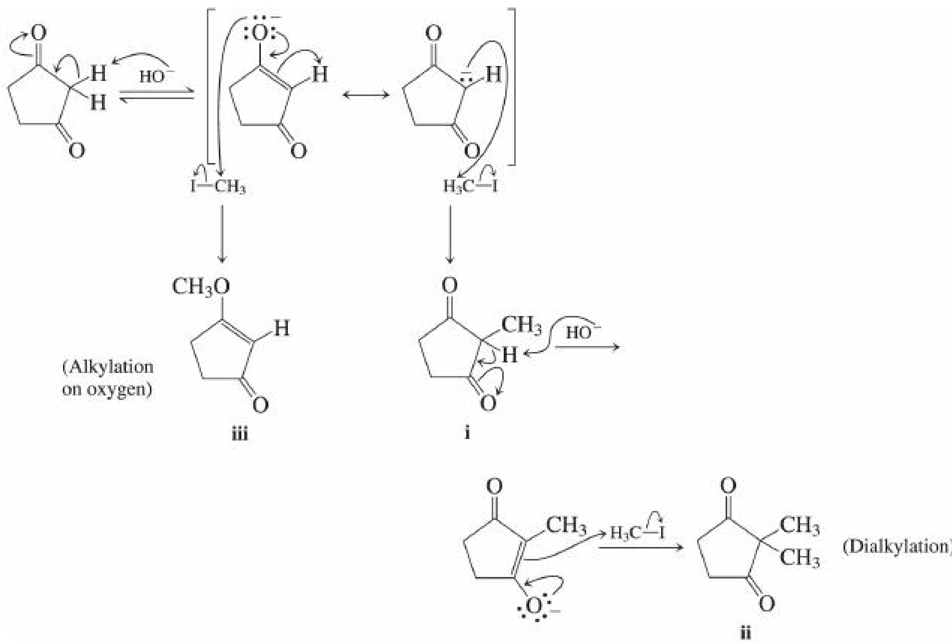
(g) B+H2,Pd—C→CH3CH2OHA (Ethanol is not a reaction participant; it is merely acommon solvent for catalytic hydrogenation.)

CH_3CH_2OH

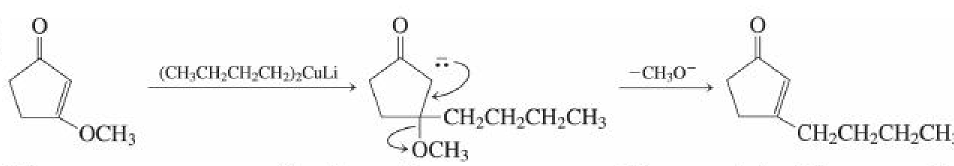
$B + H_2, Pd—C \longrightarrow A$ (Ethanol is not a reaction participant; it is merely a common solvent for catalytic hydrogenation.)

52.

(a)



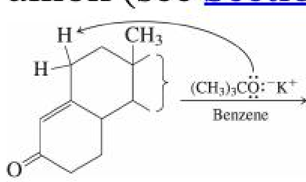
(b)

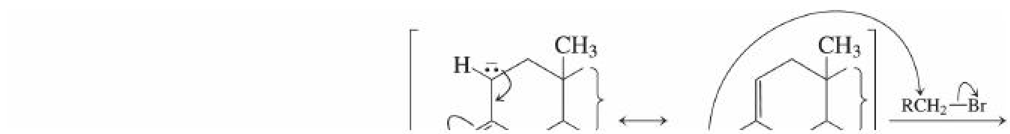


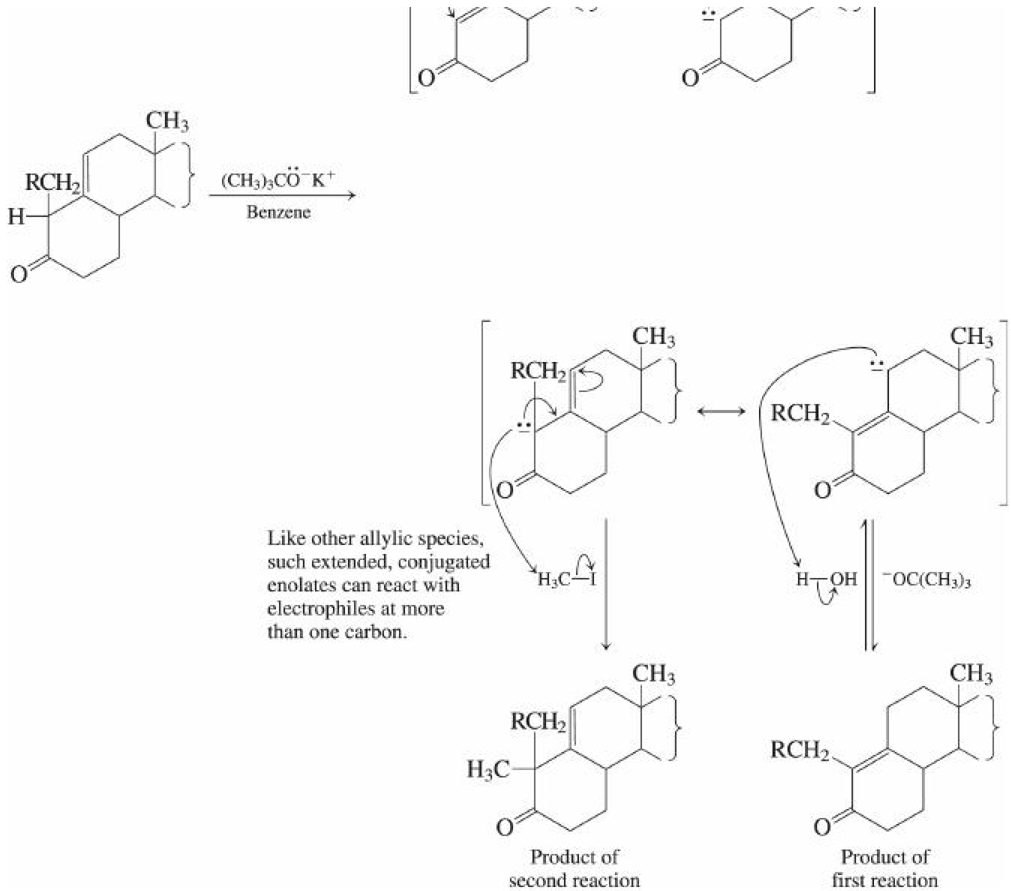
The presence of a leaving group adjacent to the enolate diverts the reaction to elimination.

53.

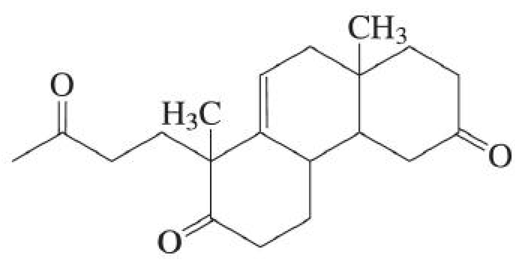
(a) Deprotonate “allylic” to give a conjugate enolatelike anion (see Section 18-11):



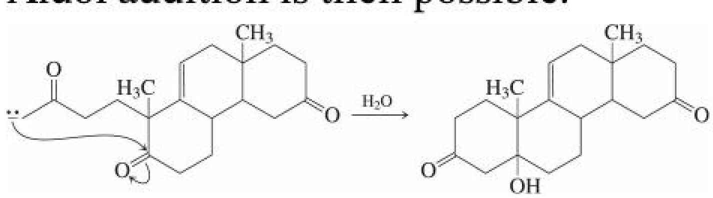




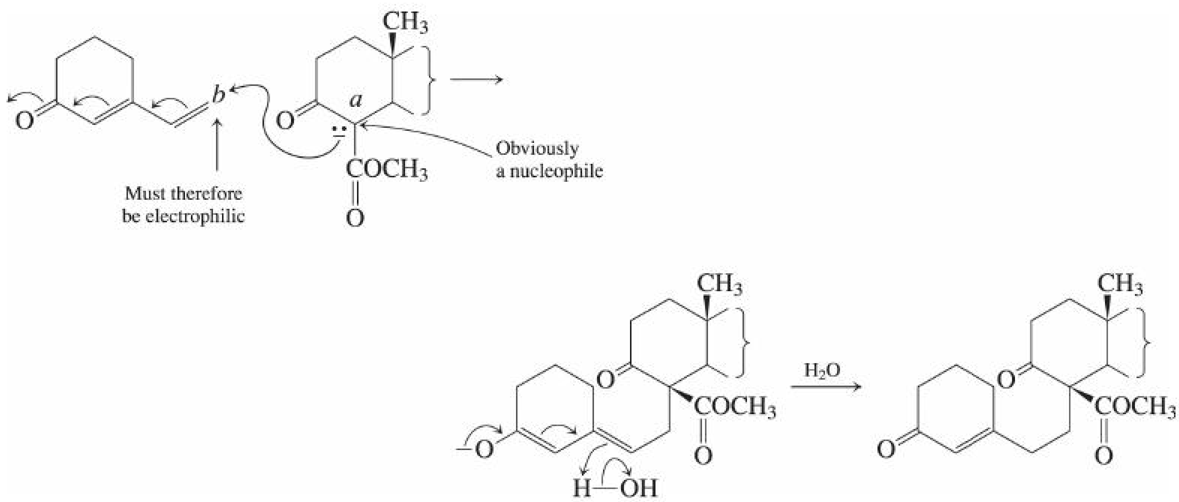
(b) H+,H2OH^+, H_2O followed by Cr6+Cr^{6+} oxidation gives



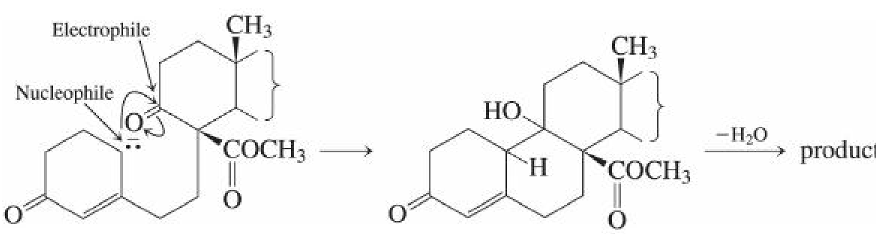
Aldol addition is then possible:



54. Analyze the bonds that have been formed in each step. Can you identify the nucleophilic and electrophilic carbons in each? That's the key: If you can identify one of the atoms going into a new bond as a nucleophile (*a* below), then the other **must** be electrophilic (*b*). So,



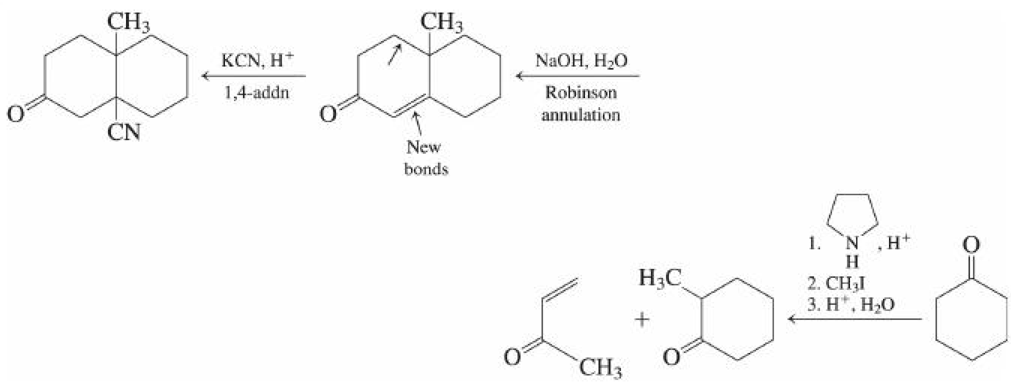
This, therefore, is a version of nucleophilic addition to an unsaturated ketone, but here the ketone is doubly (α,β,γ,δ)(α,β,γ,δ) unsaturated, and the process is a *1,6-addition.* The product enolate protonates on the δ-carbon,δ-carbon, giving an α,β-α,β-unsaturated ketone as the product. In the second reaction, remove an allylic proton with base to give an *extended* enolate.

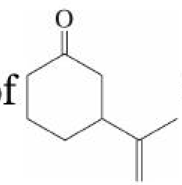


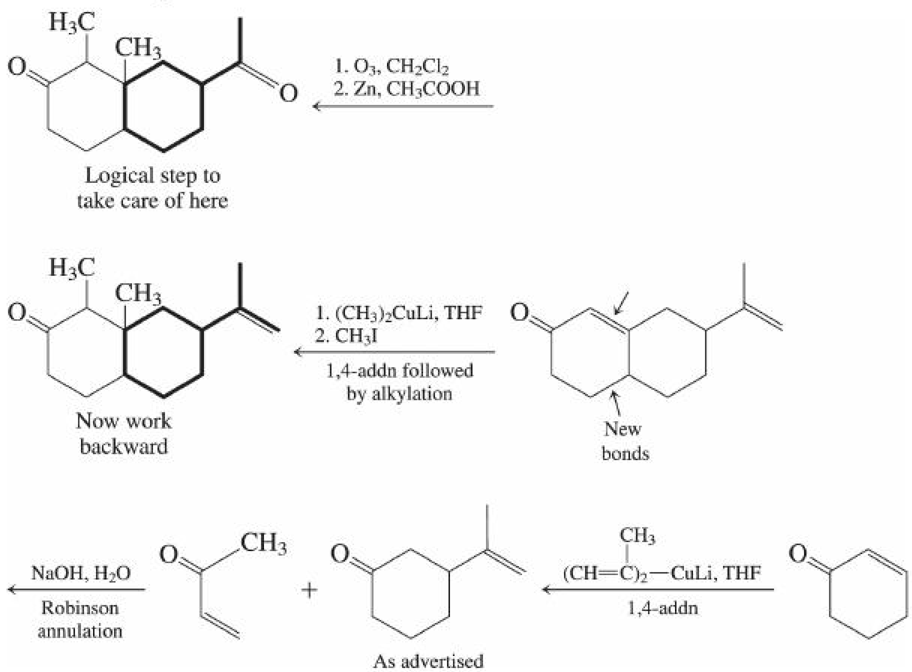
This is just a version of aldol condensation, where the nucleophile is the extended enolate ion at the γ-carbonγ-carbon of an α,β-α,β-unsaturated ketone, instead of the simple enolate at the α-carbonα-carbon of a saturated ketone.

55. Think retrosynthetically and work backwards.

(a)



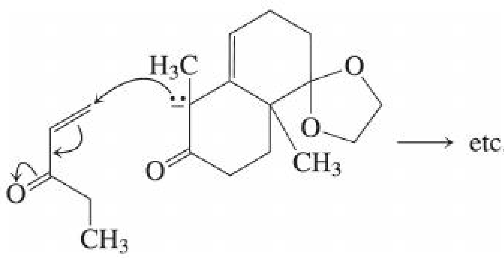
(b) Use the **hint!** Find the carbons of  in your final target molecule.



56.

(a) $CH_3CH_2\overset{\|}{C}CH{=}CH_2$, NaOH, H_2O (Robinson annulation)

(b) Exactly the same reagents as (a), but here the nucleophile is the α-carbonα-carbon of the extended enolate formed by allylic deprotonation of the α,β-α,β-unsaturated ketone:



(c) CH3ICH_3I

(d) 1. H2,Pd (reduces C=C bond),H_2, Pd (reduces Câ C bond),

2. NaBH4 (reduces C=O),$NaBH_4$ (reduces Câ O),

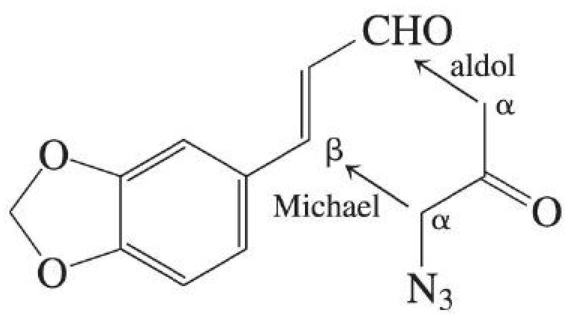
3. H+,H2O (hydrolyzes acetal),
H^+ , H_2O (hydrolyzes acetal),

4. $CH_3\overset{O}{\overset{\|}{C}}Cl$ (makes ester of newly formed alcohol)

(e) 1. Cl2,CH3COOHCl_2, CH_3COOH (chlorinates αα to ketone),

2. K2CO3,H2OK_2CO_3, H_2O (eliminates HClHCl to make α,β-α,β-unsaturated ketone)

57. Begin by “mapping” the structures of the two starting compounds onto the skeleton of the product. In this way you can find out where every atom ended up. The new six-membered ring is made by (1) connecting the azide-containing α carbonα carbon of the azidoketone with the β carbonβ carbon of the enal (a Michael addition) and (2) connecting the other α carbonα carbon of the azidoketone with the carbonyl carbon of the enal (an aldol reaction):



You’ve had plenty of opportunities to practice these two mechanisms, so hopefully you can write them out yourself by now, at least using a simple base like hydroxide. In the actual example, the use of a specialized optically pure chiral base allows formation of one stereoisomer of the product in almost enantiomerically pure form. Overall, these two processes constitute just another example of the Robinson annulation.

58. Kinetically, the bulky base will prefer to deprotonate the ketone at the less hindered, unsubstituted α-carbonα-carbon to the left of the carbonyl group. Thermodynamically, however, the more stable enolate ion will be the one on the right, containing a tetrasubstituted double bond (Section 11-7). Under condition B, the excess ketone that is present can reversibly protonate enolate ions, providing a mechanism for enolate equilibration. Under condition A, excess base is always present, and therefore there is never a significant

concentration of neutral ketone around to allow enolate equilibration to occur. Your potential energy diagram should show a lower activation barrier leading to the less-stable, higher energy "kinetic" enolate (the product on the left in the problem).

69. **(b)**

70. **(c)**

71. **(d)**

72. **(c)** is conjugated

19
Carboxylic Acids

The carboxylic acids and their derivatives contain a carbonyl group as one primary source of reactivity and therefore have a lot in common with aldehydes and ketones. However, two major features make carboxylic acids different from aldehydes and ketones. First, the HOHO proton is made much more acidic by the neighboring C=OCâ O function. Second, the HOHO may behave as a leaving group under the right conditions. You will discover the importance of this ability when you study nucleophilic additions to the carboxy group.

Outline of the Chapter

19-1 Naming the Carboxylic Acids

19-2, 19-3 Physical Properties of Carboxylic Acids

Very strongly hydrogen-bonding molecules; spectroscopy.

19-4 Acidity and Basicity of Carboxylic Acids

An analysis of resonance and inductive effects.

19-5, 19-6 Preparation of Carboxylic Acids

Reviewing some reactions you've seen before.

19-7 Reactivity of the Carboxy Group: Addition and

Keys to the Chapter

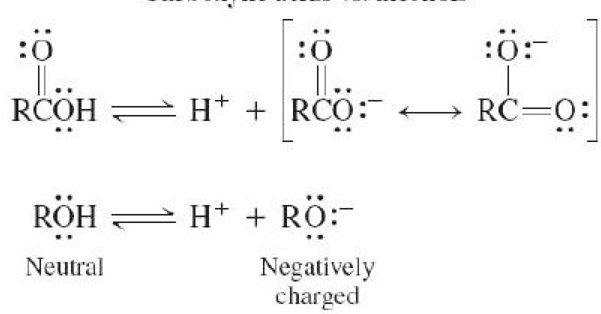
19-1 through 19-3. Nomenclature and Physical Properties of Carboxylic Acids

The one characteristic that is unique to carboxylic acids is the strong tendency to form hydrogen-bonded dimers, a process resulting in much higher melting and boiling points relative to comparable compounds (Table 19-2). Notice also the very low field position of the COOHCOOH proton in the NMR.NMR. It is very deshielded.

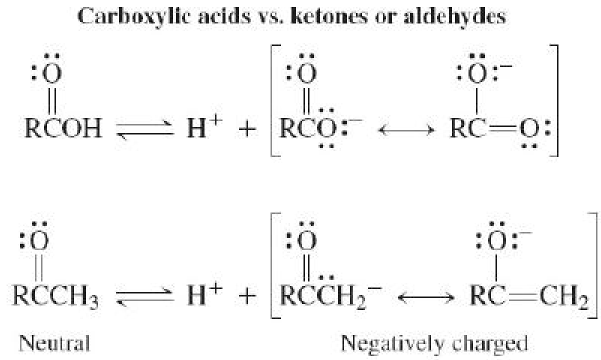
19-4. Acidity and Basicity of Carboxylic Acids

Recall that acid and base strengths are determined by the charge-stabilizing ability in the structures of the acid-base conjugate pair. There are three points to consider in evaluating the stability of a charged species: (1) the electronegativity of the charged atom(s), (2) the size of the charged atom(s), (3) stabilization of charge by either inductive effects or resonance. For carboxylic acids, the text sets up two comparisons.

Carboxylic acids vs. alcohols



Both negatively charged conjugate bases have the same negative atom (O),(O), but the carboxylate ion is stabilized by inductive effects and resonance. Because the carboxylate conjugate base is more stable, the carboxylic acid is a stronger acid than the alcohol.



Both negatively charged conjugate bases are stabilized by inductive effects of a δ+δ^+ carbonyl carbon. Both are also stabilized by resonance, but the carboxylate ion distributes its negative charge between **two** electronegative oxygens, whereas the enolate distributes its charge between a carbon and only one electronegative oxygen. The carboxylate ion is more stable, so the carboxylic acid is more acidic.

Similar analyses can be applied to base strength as well. See chapter-end Problems 30 and 49 and their answers for examples and explanations.

19-6. Preparation of Carboxylic Acids

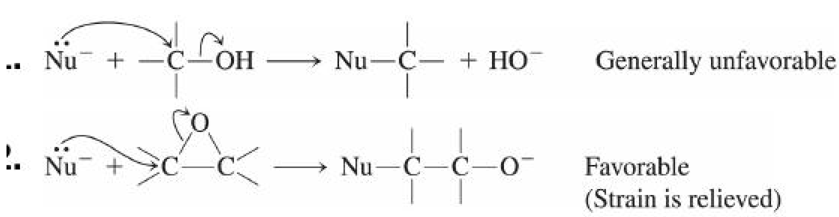
Even though the most obvious syntheses of carboxylic acids are simple functional group interconversions (mainly oxidations), there are quite a few that involve breaking or forming carbon–carbon bonds. You might take the time to do a quick review and then begin organizing these synthetic methods into appropriate categories.

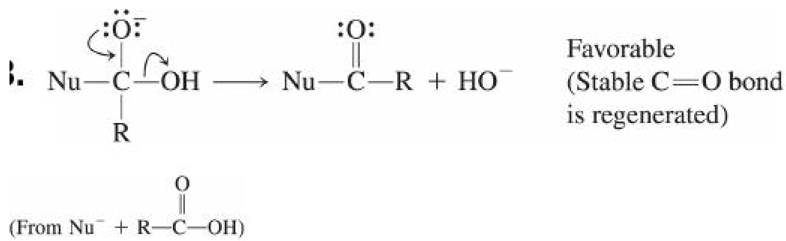
19-7. Reactivity of the Carboxy Group: Addition and Elimination

The mechanistic discussion at the start of this section is central to both the chemistry of carboxylic acids and the chemistry of their derivatives. Read it **very** carefully, and copy down some of the schemes on your own, simply to have the practice of writing them down. Here are two key points to keep in mind.

1. Carboxylic acids and their derivatives contain *potential leaving groups* attached to their carbonyl carbon. After a nucleophile adds, the leaving group may leave, giving a net substitution reaction overall. This is a two-step, **addition–elimination,** mechanism. It is *not* the same as either an SN2S_N2 or an SN1S_N1 reaction, mechanistically: SN1S_N1 and SN2S_N2 reactions *do not apply to* sp2sp^2 *hybridized carbons.*
2. For carboxylic acids, this new type of substitution process occurs best with weakly basic nucleophiles in the presence of an acid catalyst. Strong bases will deprotonate the acid faster than nucleophilic addition can take place. Therefore, if the nucleophile is a strong base and deprotonation is essentially irreversible, nucleophilic addition will be *very difficult* and will occur only with extraordinarily powerful reagents, such as LiAlH4.$LiAlH_4$.

In some of these reactions you will see some unexpected species act as leaving groups, including strong bases such as HO−HO^- and RO−.RO^-. Although these were generally far too basic to be leaving groups in ordinary SN2S_N2 reactions (Chapter 6), they can function that way here, because the tetrahedral intermediate is relatively high in energy compared with the carbonyl product, which has a very strong, very stable carbon–oxygen double bond. So, even HO−HO^- and RO−RO^- can leave in the elimination step and still result in an energetically favorable process overall. You saw a similar effect in the nucleophilic ring-opening of oxacyclopropanes with bases (Chapter 9). The high energy of the strained three-membered ring ether made expulsion of an alkoxide possible. To summarize,





Note how acid catalysis can be beneficial to all these reactions: Adding a proton to each of these oxygens before it leaves should help by converting it from a strongly basic alkoxide (i.e., bad) leaving group to a neutral alcohol or water (weakly basic, good leaving group).

19-8, 19-9, and 19-10. Acyl Halides, Anhydrides, Esters, and Amides

In these sections the general mechanisms of the previous section will be applied in illustrating the transformations of carboxylic acids into their four most important derivatives. For later synthetic applications, carefully note the reagents involved in each case. To understand these reactions, pick out a few whose mechanisms are not given in detail in the text and try to write them out, step-by-step. The practice will prove to be worthwhile.

19-11. Nucleophilic Attack at a Carboxylate Group

Strongly basic nucleophiles irreversibly deprotonate carboxylic acids, forming carboxylate anions. Addition–elimination reactions on carboxylate anions are hard to do because (1) the addition is hard to do and (2) the elimination is hard to do. More specifically, (1) it is difficult to add one anion to another due to electrostatic repulsion and (2) oxide anions are extremely bad leaving groups. Nevertheless, LiAlH4$\mathrm{LiAlH_4}$ is capable of addition to RCOO−,$\mathrm{RCOO^-}$, and the product of addition can go on to eliminate, formally, an aluminum oxide anion leaving group. The product of an acid+LiAlH4$\mathrm{acid + LiAlH_4}$ is a primary alcohol.

19-12. α-Brominationî±Bromination

A useful method of α-substitutionα-substitution in carboxylic acids is via the α-bromoα-bromo derivative, synthesized using the Hell-Volhard-Zelinsky reaction. The bromine may then serve as a leaving group in a subsequent nucleophilic substitution process to give a variety of products, including α-amino acidsα-amino acids (See Section 26-2).